

Extra Unit C practice 3

The problems that follow are examples of questions I've asked on assessments in past semesters. They are meant to give you a sense of the style of problems on assessments, but not necessarily the content of the problems. For example, just because a particular concept doesn't appear in these practice problems doesn't mean that you aren't responsible for knowing it for our assessments. You're responsible for knowing anything in the notes, the past final worksheets, and the Brightspace problems. The formulas below will be provided on Opportunity C.

Formulas

- Integral formulas

- $\int cf(x)dx = c \int f(x)dx$
- $\int [f(x) \pm g(x)]dx = \int f(x)dx \pm \int g(x)dx$
- $\int kdx = kx + C$
- $\int x^n dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1)$
- $\int \sin(x)dx = -\cos(x) + C$
- $\int \cos(x)dx = \sin(x) + C$
- $\int \sec^2(x)dx = \tan(x) + C$
- $\int \csc^2(x)dx = -\cot(x) + C$
- $\int \sec(x)\tan(x)dx = \sec(x) + C$
- $\int \csc(x)\cot(x)dx = -\csc(x) + C$
- $\int e^x dx = e^x + C$
- $\int \frac{1}{x} dx = \ln|x| + C$
- $\int f(g(x))g'(x)dx = f(g(x)) + C$

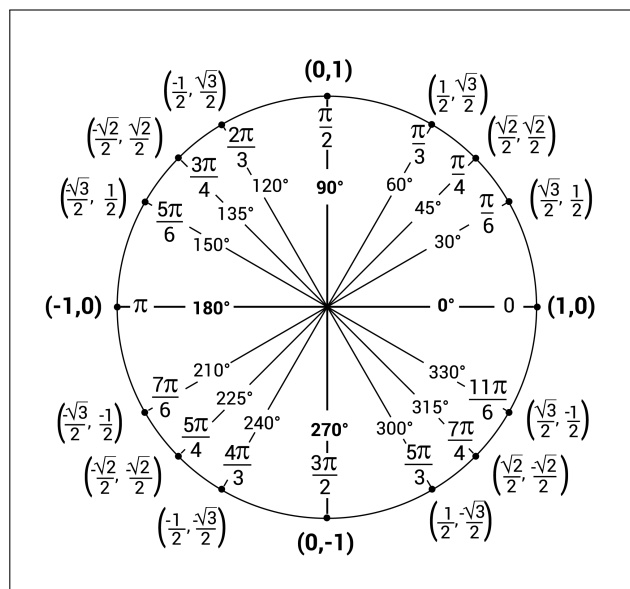
- $(f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))}$

- Fundamental Theorem of Calculus

- $\int_a^b f(x)dx = F(b) - F(a)$
- $\frac{d}{dx} \left(\int_a^x f(t)dt \right) = f(x)$

- Law of natural growth/decay

- $Q'(t) = kQ(t)$
- $Q(t) = Q_0 e^{kt}$
- $T'(t) = k(T(t) - T_s)$



C1 - The integral

- (a) The function $f(t)$ whose graph is to the right is a piecewise function consisting of a line and a semi-circle. Define a function

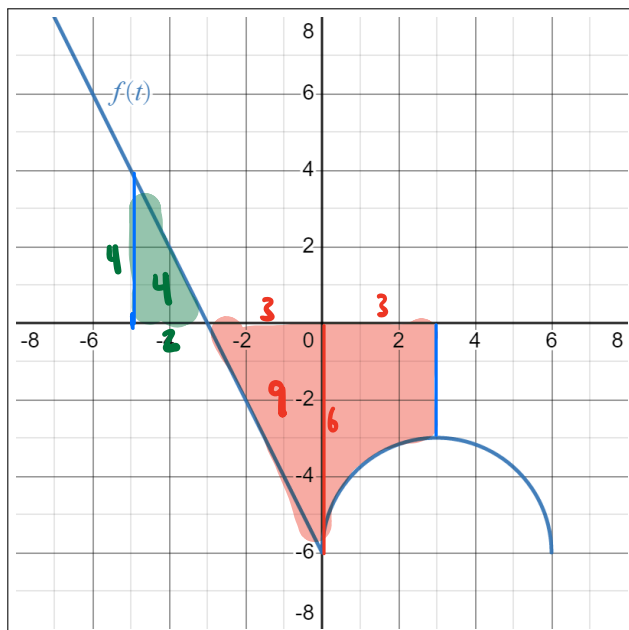
$$F(x) = \int_{x+1}^{-7x+9} f(t)dt$$

Use the graph to evaluate each of the following:

$$(i) \quad F(1) = \int_2^2 f(t) dt = 0$$

$$(ii) \quad F(2) = \int_3^{-5} f(t) dt = - \int_{-5}^3 f(t) dt$$

$$= (\triangle - \nabla - \square) = - \left(4 - 9 - \left(18 - \frac{9\pi}{4} \right) \right)$$

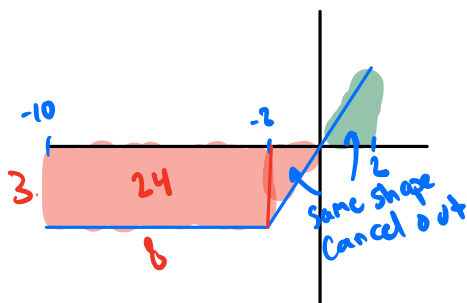


- (b) A piecewise linear function is given by

$$f(x) = \begin{cases} -3 & \text{if } x \leq -2 \\ \frac{3x}{2} & \text{if } -2 \leq x \end{cases}$$

Evaluate the following integral by interpreting it in terms of areas. Include a sketch of the areas you use.

$$\int_{-10}^2 f(x) dx$$



$$= -24$$

C2 - The Fundamental Theorem of Calculus

(a) Consider the function

$$f(x) = \int_{2x}^{3x+1} \sin\left(\frac{\pi t^4}{64}\right) dt = \int_0^{3x+1} \sin\left(\frac{\pi t^4}{64}\right) dt - \int_0^{2x} \sin\left(\frac{\pi t^4}{64}\right) dt$$

Find the linearization of $f(x)$ at $x = -1$ and use it to approximate $f(-1.1)$.

$$f'(x) = \sin\left(\frac{\pi(3x+1)^4}{64}\right) \cdot 3 - \sin\left(\frac{\pi(2x)^4}{64}\right) \cdot 2$$

$$f'(-1) = \sin\left(\frac{\pi}{4}\right) \cdot 3 - \sin\left(\frac{\pi}{4}\right) \cdot 2 = \frac{\sqrt{2}}{2}$$

$$f(-1) = \int_{-2}^{-1} \sin\left(\frac{\pi t^4}{64}\right) dt = 0$$

$$L(x) = \frac{\sqrt{2}}{2}(x - (-1)) + 0$$

$$f(-1.1) \approx L(-1.1) = \frac{\sqrt{2}}{2}(-1.1 - (-1)) = \boxed{-\frac{\sqrt{2}}{20}}$$

(b) If $g(x) = (\sqrt[4]{x} + 1)^2 \cdot (e^x - 1) \cdot (\cos(\pi x) + 1)$, find the signed area between the x -axis and the *derivative* of $g(x)$ on the interval $[0, 1]$. In other words, compute the following integral.

$$\int_0^1 g'(x) dx$$

$$\int_0^1 g'(x) dx = g(1) - g(0)$$

$$= (\sqrt[4]{1} + 1)^2 \cdot (e^1 - 1) \cdot (\cos(\pi \cdot 1) + 1) - (\sqrt[4]{0} + 1)^2 \cdot (e^0 - 1) \cdot (\cos(\pi \cdot 0) + 1)$$

$$= 0 - 0 = \boxed{0}$$

C3 - Integration by substitution

Find the following integrals. Clearly show each step of your work for full credit.

$$(a) \int_e^6 \frac{dx}{x \ln x}$$

$$\begin{aligned} u &= \ln x & u(e) &= \ln e = 1 \\ du &= \frac{1}{x} dx & u(6) &= \ln 6 \\ x du &= dx \end{aligned}$$

$$\int_1^{\ln 6} \frac{x du}{x \cdot u} = \int_1^{\ln 6} u^{-1} du = \ln|u| \Big|_1^{\ln 6} = \ln(\ln(6)) - \ln(1)$$
$$\boxed{= \ln(\ln(6))}$$

$$(b) \int \frac{x^3}{\sqrt{x^2+1}} dx$$

$$u = x^2 + 1 \rightarrow x^2 = u - 1$$

$$du = 2x dx$$

$$\frac{du}{2x} = dx$$

$$\int \frac{x^3}{\sqrt{u}} \cdot \frac{du}{2x} = \frac{1}{2} \int \frac{x^2}{\sqrt{u}} du = \frac{1}{2} \int \frac{u-1}{\sqrt{u}} du = \frac{1}{2} \int u^{\frac{1}{2}} - u^{-\frac{1}{2}} du = \frac{1}{2} \left(\frac{2}{3} u^{\frac{3}{2}} - 2u^{\frac{1}{2}} \right) + C$$

$$\boxed{= \frac{1}{3} (x^2+1)^{\frac{3}{2}} - (x^2+1)^{\frac{1}{2}} + C}$$

C4 - Inverse functions and their derivatives

Let $f(x) = 2x + \sqrt{x^3 + 1}$

(a) Show that $f(x)$ is one-to-one on its domain.

$$x^3 + 1 \geq 0 \rightarrow x \geq -1$$

$$f'(x) = 2 + \frac{1}{2}(x^3 + 1)^{-\frac{1}{2}} \cdot 3x^2 \geq 2 \quad \text{for } x \geq -1$$

So $f(x)$ is always increasing, so it is one-to-one.

(b) Find the derivative of $f^{-1}(x)$ at $x = 1$.

$$(f^{-1})'(1) = \frac{1}{f'(f^{-1}(1))} \quad f(0) = 2 \cdot 0 + \sqrt{0^3 + 1} = 1 \quad \text{so } f^{-1}(1) = 0$$

$$= \frac{1}{f'(0)} = \frac{1}{2 + \frac{1}{2}(0^3 + 1)^{-\frac{1}{2}} \cdot 3 \cdot 0^2} = \boxed{\frac{1}{2}}$$

C5 - Exponential growth and decay

Bacteria grow in a Petri dish (namely a dish containing a solution of nutrients) in such a way that at any moment the rate of change of the area of the dish covered by the bacteria is proportional to the area of the dish covered by the bacteria. Exactly 1 day after the bacteria begin to grow, they cover 6 square millimeters of the petri dish, and at that moment are growing at a rate of $6 \cdot \ln(1.5)$ square millimeters per day.

- (a) Write a function $A(t)$ for the area of the dish covered by the bacteria t days after they start to grow.

$$A'(t) = k \cdot A(t)$$

$$6 \cdot \ln(1.5) = k \cdot 6 \rightarrow k = \ln(1.5)$$

$$A(t) = A_0 \cdot e^{\ln(1.5)t} = A_0 \cdot 1.5^t$$

$$A(1) = A_0 \cdot 1.5^1 = 6 \rightarrow A_0 = 4$$

$$A(t) = 4 \cdot (1.5)^t$$

- (b) How fast is the area of the bacteria changing at $t = 2$ days?

$$A'(2) = \ln(1.5) \cdot A(2)$$

$$= \ln(1.5) \cdot 4 \cdot (1.5)^2$$